

MCP Playground : Custom MCP & Official MCP Implementation

1. No Code Playwright MCP automation: (Create CRM lead in LeafTaps Application):

The screenshot displays the MCP Playground interface. On the left, the 'EXTENSIONS' sidebar shows 'Playwright' installed. The main panel shows the 'Playwright MCP' details, including its identifier 'microsoft/playwright-mcp' and version '0.0.1-seed'. The 'OUTPUT' tab shows the server starting and discovering 24 tools. On the right, the 'CHAT' panel shows a successful automation run for 'RUN IN PLAYWRIGHT WITH CHROME BROWSER: LOGIN TO ...'. The chat log includes steps like 'Navigate to login page', 'Enter credentials and login', 'Click CRM/SFA link', 'Navigate to Create Lead form', and 'Fill in lead details and submit'. A table of lead details is shown:

Field	Value
Company Name	Testleaf (11655)
First Name	Prabakaran
Last Name	Murugan
Department	Agentic AI Developer
Email Address	Test-AI@demo.com

The chat also confirms: 'The lead has been successfully created. The page redirected to View Lead (partyId=11655) confirming the submission. Here's a summary of what was entered and verified on the confirmation page:'. A 'Todos' list at the bottom right shows the completed steps.

2. Local Mongo MCP: Fetch User Story and create Test Cases for the same:

The screenshot displays the MCP Playground interface with the 'MongoDB MCP' extension installed. The main panel shows the 'MongoDB MCP' configuration, including the name 'mongodb', type 'stdio', and command 'npx'. The 'OUTPUT' tab shows the server starting and discovering 18 tools. On the right, the 'CHAT' panel shows a successful automation run for 'LIST MONGODB DATABASES'. The chat log includes steps like 'Navigate to login page', 'Enter credentials and login', 'Click CRM/SFA link', 'Navigate to Create Lead form', and 'Fill in lead details and submit'. A table of lead details is shown:

Field	Value
Company Name	Testleaf (11655)
First Name	Prabakaran
Last Name	Murugan
Department	Agentic AI Developer
Email Address	Test-AI@demo.com

The chat also confirms: 'The lead has been successfully created. The page redirected to View Lead (partyId=11655) confirming the submission. Here's a summary of what was entered and verified on the confirmation page:'. A 'Todos' list at the bottom right shows the completed steps.

Custom Remote MCP for Postman:

Model Provider

☒ Groq ☐ OpenAI

Jira MCP

☒ Local STUDIO ☐ Remote HTTP

MongoDB MCP

☒ Local STUDIO

Postman MCP

☐ Local STUDIO ☒ Remote HTTP

Connection Visualization

React Frontend → Express Backend → MCP Connection Manager → Local STUDIO / Remote HTTP

Jira

stdio

Not started

Connect (local)

MongoDB

stdio

Not started

Connect (local)

Postman

http

Connected (4 tools)

Disconnect

Prompt

Fetch me Collections name in postman

Send

Response

The collections names in postman are: 1. GenAI-May 2. GenAI-May-Home Work 3. LangflowAPI 4. LangflowAPI_Customized Fields 5. LangflowAPI_Customized Fields_USReviewer 6. Moderation Layer_Architecture.md 7. RAG MongoDB - Complete Ingestion & Retrieval 8. RAG MongoDB - Data Ingestion 9. Restful Booker API 10. Test Case Generator RAG - Backend 11. jiraTest Copy 12. jiraTest_Langflow

Tool Trace Timeline

1 round · 1 call · 2705ms total · Model finished without needing more tools

1 postman_get_collections postman · remote/http

✓ success · 838ms

Error Simulation

Fire known-bad requests at the real backend to see how each error is handled.

Call a tool that isn't approved Run scenario

Custom Remote MCP for Jira:

Model Provider

☒ Groq ☐ OpenAI

Jira MCP

☐ Local STUDIO ☒ Remote HTTP

MongoDB MCP

☒ Local STUDIO

Postman MCP

☒ Local STUDIO ☐ Remote HTTP

Connection Visualization

React Frontend → Express Backend → MCP Connection Manager → Local STUDIO / Remote HTTP

Jira

http

Connected (1 tools)

Disconnect

MongoDB

stdio

Connected (3 tools)

Disconnect

Postman

http

Connected (4 tools)

Disconnect

Prompt

Fetch me two User stories from GenAI_UserStories_Jira workspace in jira

Send

Response

The two user stories from the GenAI_UserStories_Jira workspace in Jira are: 1. GUJ-25: Employee Profile Employee Self, assigned to Ajay AS, with a status of To Do. 2. GUJ-24: Ahc Reports Reports Ahc, unassigned, with a status of To Do.

Tool Trace Timeline

1 round · 1 call · 2506ms total · Model finished without needing more tools

1 jira_get_stories jira · remote/http

✓ success · 834ms

Error Simulation

Fire known-bad requests at the real backend to see how each error is handled.

Call a tool that isn't approved Run scenario

Custom Local MCP for Mongoddb:

Compare Local MCP and Remote MCP across Jira, MongoDB, and Postman.

Model Provider

Groq

OpenAI

Jira MCP

Local STUDIO

Remote HTTP

MongoDB MCP

Local STUDIO

Postman MCP

Local STUDIO

Remote HTTP

Connection Visualization

React Frontend → Express Backend → MCP Connection Manager → Local STUDIO / Remote HTTP

Jira

studio

Connected (1 tools)

Disconnect

MongoDB

studio

Connected (3 tools)

Disconnect

Postman

studio

Connected (4 tools)

Prompt

fetch the databases in mongo

Send

Response

The databases available in your MongoDB instance are: 1. kb_TstGenAI 2. admin 3. local

Tool Trace Timeline

1 round · 1 call · 1463ms total · Model finished without needing more tools

1 mongo_get_databases mongo · local/stdio

✓ success · 726ms

Error Simulation

Fire known-bad requests at the real backend to see how each error is handled.

Call a tool that isn't approved

Run scenario

Custom Local MCP for Postman:

Compare Local MCP and Remote MCP across Jira, MongoDB, and Postman.

Model Provider

Groq

OpenAI

Jira MCP

Local STUDIO

Remote HTTP

MongoDB MCP

Local STUDIO

Postman MCP

Local STUDIO

Remote HTTP

Connection Visualization

React Frontend → Express Backend → MCP Connection Manager → Local STUDIO / Remote HTTP

Jira

studio

Connected (1 tools)

Disconnect

MongoDB

studio

Connected (3 tools)

Disconnect

Postman

studio

Connected (4 tools)

Disconnect

Prompt

List out all the collections name in my postman

Send

Response

Here is the list of collection names in your Postman: 1. GenAI-May 2. GenAI-May-Home Work 3. LangflowAPI 4. LangflowAPI_Customized Fields 5. LangflowAPI_Customized Fields_USReviewer 6. Moderation Layer_Architecture.md 7. RAG MongoDB - Complete Ingestion & Retrieval 8. RAG MongoDB - Data Ingestion 9. Restful Booker API 10. Test Case Generator RAG - Backend 11. jiraTest Copy 12. jiraTest_Langflow

Tool Trace Timeline

1 round · 1 call · 1934ms total · Model finished without needing more tools

1 postman_get_collections postman · local/stdio

✓ success · 587ms

Error Simulation

Fire known-bad requests at the real backend to see how each error is handled.

MCP Playground : Custom MCP & Official MCP Implementation

Custom Local MCP for Jira:

MCP Playground

Compare Local MCP and Remote MCP across Jira, MongoDB, and Postman.

Model Provider

☒ Groq ☐ OpenAI

Jira MCP

☒ Local STUDIO ☐ Remote HTTP

MongoDB MCP

☒ Local STUDIO

Postman MCP

☒ Local STUDIO ☐ Remote HTTP

Connection Visualization

React Frontend → Express Backend → MCP Connection Manager → Local STUDIO / Remote HTTP

Jira

stdio

Connected (1 tools)

Disconnect

MongoDB

stdio

Connected (3 tools)

Disconnect

Prompt

Fetch me two user stories from my "GenAI_UserStories_Jira" jira workspace

Send

Response

Here are two user stories from your "GenAI_UserStories_Jira" Jira workspace: 1. Key: GUJ-25, Summary: Employee Profile Employee Self, Status: To Do, Assignee: Ajay AS 2. Key: GUJ-24, Summary: AHC Reports Reports AHC, Status: To Do, Assignee: Unassigned These user stories are currently in the "To Do" status and are assigned to Ajay AS and Unassigned, respectively.

Tool Trace Timeline

1 round · 1 call · 1526ms total · Model finished without needing more tools

1 jira_get_stories jira · local/stdio
✓ success · 698ms

Error Simulation

Architecture Plan and vibe coding implementation of the Official Postman MCP:

EXPLORER

OPEN EDITORS

architecture.md

MCP-PLAYGROUND

backend

src

.env

package.json

docs

frontend

mcp

node_modules

.env

.gitignore

architecture.md

package-lock.json

package.json

README.md

architecture.md

Architecture Plan: "Official Postman" Mode for MCP Playground

1 # Architecture Plan: "Official Postman" Mode for MCP Playground

2

3 ## Overview

4

5 Add a third radio-button option - **"Official Postman"** - inside the

6 ****Postman MCP**** panel in the UI.

7 When selected, a ****toolset selector**** (Minimal / Code / Full / Learn)

8 appears.

9 The backend then spawns ``npx @postman/postman-mcp-server [--flag]`` as

10 a local STUDIO subprocess (identical transport to the existing "Local

11 STUDIO" path), authenticated via ``POSTMAN_API_KEY``, and exposes ****all**

12 **100+ tools**** from the official Postman MCP server to the agent loop.

13

14 **### Why STUDIO (npx) and not ``https://mcp.postman.com``**

15

16 The official remote endpoint (``https://mcp.postman.com``) uses OAuth as

17 its primary auth method.

18 API-key-over-Bearer is available for EU only.

19 The local npx package uses ``POSTMAN_API_KEY`` directly and is the

20 recommended path for programmatic integration without an OAuth flow.

21 ``POSTMAN_API_KEY`` is already tracked in ``backend/.env`` via ``env``.

22 ``postman.apiKey``.

23

24 **### Connection data flow**

25

26 ...

27

28 UI (radio: "official" + toolset dropdown)

29 → usePlayground sends { connections: { postman: "official" },

30 postmanToolset: "minimal" }

31

32 → apiClient.sendChat / apiClient.connect

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34 → Express backend (connection.controller / playground.controller)

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36 → MCPConnectionManager.connect("postman", "official", {

37 toolset: "minimal" })

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PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

ajayuchiha@Ajays-MacBook-Air mcp-playground % JIRA_REMOTE_MCP_TOKEN=ATA

TT3xFfGF0yvp52aq7ojYby1rUcVMy9k6RAHyMbG029ix26XEGE7Tk8d1hvfHLuqZlp5FKmA

XX4b7G2g8jD7aJNpRwI-uhLwLBK6kfa1t9Zie6JSXfIEQVwDcaH90ynb8XlqhAFRK59bdy

Ohmitw_8nAQ775PyOkagG2Pt5GGrf6JtG_XL00=A8870E53 npm run mcp:jira:remote

Jira remote MCP endpoint listening on http://localhost:4001/mcp

ajayuchiha@Ajays-MacBook-Air mcp-playground % npm run mcp:jira:remote

> mcp-playground@1.0.0 mcp:jira:remote

> node mcp/jira/remote/server.js

Jira remote MCP endpoint listening on http://localhost:4001/mcp

CHAT

CLAUDE CODE

Update

ADD "OFFICIAL POSTMAN" OPTION PLAN

dropdown when official mode is

active for Postman

Added postmanToolset state

includes it in sendChat

payload

sendChat forwards

postmanToolset ; connect

forwards toolset

connect() accepts and pass

toolset

Wires postmanToolset /

setPostmanToolset /

POSTMAN_TOOLSETS props

to ConnectionModeRadioGroup

for Postman; passes

postmanToolset to

ConnectionStatus

Accepts postmanToolset ,

passes it as toolset to

connect() for the Postman

system

Todos (17/17)

Step 1: Update backend/.env

Step 2: Extend environment.js

Step 3: Update mcp.config.js

Step 4: Update toolRegistry.js

Step 5: Verify localStudio.client.js

Step 6: Update mcpClient.factory.js

architecture.md

Describe what to build

Agent Claude Sonnet 4.6 High 1M

Local Default permissions (sandboxed)

Official Postman MCP Implementation with toolset dropdown:

localhost:5173

Jira MCP

Local STUDIO

Remote HTTP

MongoDB MCP

Local STUDIO

Postman MCP

Local STUDIO

Remote HTTP

Official Postman

Toolset:

Minimal

Code

Full

Learn

Essential tools — works

Connection Manager

React Frontend → Express Backend → MCP

Connection Manager → Local STUDIO / Remote HTTP

Jira

studio

Connected (1 tools)

Disconnect

MongoDB

studio

Connected (3 tools)

Disconnect

Postman

studio

Connected (42 tools)

Disconnect

Some Issues Faced with either model token limit or model limitations with Groq Models:

Model Provider

☒ Groq ☐ OpenAI

Jira MCP

☒ Local STDIO ☐ Remote HTTP

MongoDB MCP

☒ Local STDIO

Postman MCP

☐ Local STDIO ☐ Remote HTTP
☒ Official Postman
Toolset: Minimal

Prompt

Fetch "LangflowAPI" collection from postman

Send

Error

429 {"error":{"message":"Rate limit reached for model `llama-3.3-70b-versatile` in organization `org\_01kqwa3hgpfwm8y3kczt665xm` service tier `on\_demand` on tokens per day (TPD): Limit 100000, Used 71453, Requested 37859. Please try again in 2h14m5.567999999s. Need more tokens? Upgrade to Dev Tier today at https://console.groq.com/settings/billing","type":"tokens","code":"rate_limit_exceeded"}}
▶ Technical details

Tried with other Groq models, same issue. Local/Http Postman MCPs are working fine:

Model Provider

☒ Groq ☐ OpenAI

Jira MCP

☒ Local STDIO ☐ Remote HTTP

MongoDB MCP

☒ Local STDIO

Postman MCP

☒ Local STDIO ☐ Remote HTTP
☐ Official Postman

Prompt

Fetch any collection from postman

Send

Response

The Postman collections are: 1. GenAI-May 2. GenAI-May-Home Work 3. LangflowAPI 4. LangflowAPI_Customized Fields 5. LangflowAPI_Customized Fields_USReviewer 6. Moderation Layer_Architecture.md 7. RAG MongoDB - Complete Ingestion & Retrieval 8. RAG MongoDB - Data Ingestion 9. Restful Booker API 10. Test Case Generator RAG - Backend 11. jiraTest Copy 12. jiraTest_Langflow

Tool Trace Timeline

1 round · 1 call · 7404ms total · Model finished without needing more tools
1 postman_get_collections postman · local/stdio
✓ success · 5890ms

Error Simulation

Fire known-bad requests at the real backend to see how each error is handled.

Connection Visualization

React Frontend → Express Backend → MCP Connection Manager → Local STDIO / Remote HTTP

Jira stdio Connected (1 tools) Disconnect

MongoDB stdio Connected (3 tools) Disconnect

Postman stdio Connected (42 tools)